PFC AI Design Agent

Documentation Standards

Finalised rules for equation format, chapter splash pages,
technical verbiage, graph documentation, and table documentation

Adopted from: PFC_Design_Report_Steps1_15 and four reference documents

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This document finalises five documentation standards that were not fully defined in the previous agreement PDF. Every standard is shown with a live worked example so the Documentation Agent has an unambiguous template.

Part 1 — Chapter Splash Page

✓ AGREED One Dedicated Full Page at the Start of Every Chapter

Every chapter opens with a full-page colour panel showing only the chapter number, chapter title, the core question the chapter answers, and a brief bullet list of what the chapter covers. No calculations, no tables, no figures on this page. It acts as a visual divider and sets the reader's expectation before the first calculation appears.

1.1 What the Splash Page Contains

Element	Content	Style
Chapter label	"CHAPTER 3" — small caps above the title	Small, muted, uppercase tracking
Chapter title	Full chapter name — e.g. "PFC Inductor Sizing"	Large, bold, white
Core question	The single question the chapter answers — e.g. "What inductance and winding do we need?"	Bold white, 10 pt
Scope bullets	3–5 one-line bullets listing the key topics covered	Muted, 9.5 pt, arrow prefix
Chapter colour	Full-page background in the chapter's unique colour (agreed per chapter)	Unique per chapter

1.2 Live Example — Chapter 3 Splash Page

The following is exactly how the Chapter 3 splash page will look in the report:

CHAPTER 3

PFC Inductor Sizing

Core question: What inductance and winding do we need?

→ Target inductance derived from ripple specification at worst-case operating point

→ Per-phase currents computed at all 9 operating points

→ Ripple cancellation factor K(D) — the interleaving benefit quantified

→ Material, core geometry, and wire selection with full rationale

→ First-pass loss and thermal estimates used to make the selection decision

Each chapter has its own unique colour. Chapter 1: navy, Chapter 2: dark green, Chapter 3: dark amber, Chapter 4: dark purple, Chapter 5: dark teal, Chapter 6: dark slate.

Part 2 — Equation Format Standard

✓ AGREED Equation Format Adopted from PFC_Design_Report_Steps1_15

The equation format from the reference Word document is adopted exactly. Every calculation is broken into numbered sub-steps. Each sub-step shows the symbolic equation first, then the numerical substitution on the next line, then the result with unit and a PASS/FAIL verdict where applicable.

2.1 The Four-Line Equation Template

Every equation step uses this four-line structure:

Line	Content	Example
Line 1	**Step label + description** — what is being computed, with equation reference	Step 1 Hold-up capacitance: [Eq. 15.1]
Line 2	**Symbolic equation** — every variable as defined in the Nomenclature table	$C_{hold} = (2 \cdot P_{out} \cdot t_{hold}) / (V_{out}^2 - V_{dc,min}^2)$
Line 3	**Numerical substitution** — actual design values substituted in	$C_{hold} = (2 \times 3600 \times 0.020) / (390^2 - 290^2)$
Line 4	**Result** — value with unit; PASS/FAIL verdict if applicable	C_{hold} = 2117.6 µF → governs sizing ✓

2.2 Live Example — Complete Calculation Block

The following shows three consecutive equation steps as they will appear in the report. This example is from Chapter 3, Section 3.1.1 — Target Inductance.

3.1.1 Target Inductance Derivation

CONCEPT Why We Size at the Low-Line Full-Load Crest

The inductor is sized at 90 Vac, full load — the operating point that demands the most inductance. At low line the duty cycle is highest ($D \approx 0.67$), making the volt-second product $V_{in} \times D$ the largest. A larger volt-second product means more flux swing per switching cycle, which means more ripple current for a given inductance. If the inductance meets the ripple target here, it meets it everywhere.

Step 1 Input ripple from specification (r = 9.5% of peak input current): *Eq. 3.1*

$$\Delta I_{in,pp} = r \cdot I_{in,pk}$$

$$\Delta I_{in,pp} = 0.095 \times 28.30 \text{ A}$$

$$\boldsymbol{\Delta I_{in,pp} = 2.6885 \text{ A}}$$

Step 2 Per-phase ripple — larger than input ripple by factor 1/K(D■■): *Eq. 3.2*

$$\Delta I\_L,pp = \Delta I\_in,pp / K(D\blacksquare\blacksquare)$$

$$\Delta I\_L,pp = 2.6885 / 0.4765$$

$$\mathbf{\Delta I\_L,pp = 5.642\ A}$$

Step 3 Target per-phase inductance from volt-second balance: *Eq. 3.3*

$$L\_target = V\_in,pk \cdot D / (f\_sw \cdot \Delta I\_L,pp)$$

$$L\_target = 127.28 \times 0.6736 / (70000 \times 5.642)$$

$$\mathbf{L\_target = 216.5\ \mu H \rightarrow round\ to\ standard\ 240\ \mu H\ ✓\ ADOPT\ 240\ \mu H}$$

PITFALL Do Not Apply the Ripple Ratio Directly to the Per-Phase Current

The 9.5% ratio r applies to the *input* current, not the per-phase inductor current. The per-phase ripple is larger by the factor 1/K(D■■) because the two phases partially cancel at the input. Using ΔI_L = r × I_pk directly gives 5.642 A × 0.4765 = 2.69 A — only 48% of the correct per-phase ripple. This error would produce an inductor that is undersized by roughly 2×.

DECISION L_target = 240 μH per phase

The 216.5 μH computed value is rounded up to 240 μH — the next standard core AL produces a turns count that results in this inductance. The 10.8% margin over the computed value provides tolerance for AL uncertainty (±8%) and ensures the ripple target is met even at the AL-minimum corner.

Part 3 — Technical Verbiage (Design Interpretation)

✓ AGREED A Technical Interpretation Paragraph After Every Major Calculation Block

After every significant calculation or result — particularly at the end of a section or after a table — a "Design Interpretation" paragraph is written. It explains what the result means physically, how it affects the rest of the design, what would change if the value were different, and why the result is acceptable or concerning. This is the paragraph that makes the report educational, not just computational.

3.1 What Technical Verbiage Must Cover

- **Physical meaning:** What does this number represent in the real component or circuit?
- **Design impact:** What other parameters does this value control or constrain?
- **Sensitivity:** What happens if this value is 10–20% higher or lower?
- **Engineering assessment:** Is this a comfortable margin, a tight design, or a warning?
- **Connection to next step:** How does this feed into the next calculation?

3.2 Visual Treatment — The Interpretation Block

Technical verbiage is placed in a dedicated "Design Interpretation" block with a teal-green left border and a light background, clearly distinct from calculation steps and annotation boxes. This makes it easy to skip for expert readers but easy to find for those who need the explanation.

3.3 Live Examples

Example A — After the target inductance calculation (Section 3.1.1):

What L = 240 µH Means for the Design

The 240 µH per-phase inductance is the single most influential parameter in the entire power stage. It directly sets the inductor ripple current, which in turn determines the peak current through every semiconductor, the RMS current in the capacitor bank, the core operating point, and the winding temperature rise. A larger inductance reduces all of these stresses at the cost of a physically larger, heavier, and more expensive core. A smaller inductance increases stresses and may push the design into discontinuous conduction mode at light load. At 240 µH the peak per-phase current at low line full load is 14.15 A + 2.82 A = 16.97 A — 20% above the average crest current. This 20% overhead sets the saturation margin requirement for the core: the inductance must not degrade significantly at 16.97 A DC bias. The core selection in Section 3.4 is optimised directly against this criterion.

Example B — After the K(D) cancellation factor calculation (Section 3.2.2):

What K(D) Means for the EMI Filter and the Inductor

K(D) is the ratio of the residual input ripple after interleaving to the per-phase ripple before interleaving. At 90 Vac (D = 0.67), K = 0.477 — meaning the two phases cancel 52% of the per-phase ripple at the input terminals. At 132 Vac (D ≈ 0.5), K drops to 0.100 — cancellation reaches 90%, the best the two-phase topology can achieve. This cancellation has two direct consequences for the system design. First, the EMI differential-mode filter sees only K × ΔI_L,pp of ripple current — roughly half the ripple of a single-phase design at most operating points, allowing a smaller, cheaper filter. Second, the cancellation benefit applies only to the input current; each individual inductor still carries the full per-phase ripple. Core loss, copper loss, and thermal sizing in Sections 3.6 and Chapter 4 are all computed on the per-phase values without any K(D) reduction.

Example C — After the stability scorecard in Chapter 6 (Section 6.6):

What the Stability Scorecard Tells the Designer

A phase margin of 63° in the current loop and 80° in the voltage loop at the sizing corner places this design firmly in the "over-damped" category. In step-load response terms, these margins correspond to a damping ratio ζ ≈ 0.65 — the output voltage will overshoot by less than 3% on a load step, with no ringing. The 23.7 dB of 120 Hz rejection in the voltage loop means that the 120 Hz bus ripple modulates the current reference by less than 6.5% — well within the range where input current THD remains IEC 61000-3-2 compliant. The worst-case operating point for the voltage loop is the high-line corner (180 Vac, 3600 W) where the plant gain is highest and the RHP zero is closest to the crossover frequency. All nine points pass, confirming the compensator is robust across the full operating envelope. The 20 Hz crossover frequency of the voltage loop means the controller rejects disturbances faster than the human eye can perceive a flicker in the load — important for data-centre applications where load steps are frequent and the bus must remain stable.

Part 4 — Graph Documentation Standard

✓ AGREED Every Graph Has Four Components: Introduction, Figure, Caption, Interpretation

Graphs are first-class citizens in the report. Every graph is preceded by an introductory sentence telling the reader what to look for. It is followed by a numbered caption. After the caption, a technical interpretation paragraph explains what the graph reveals about the design.

4.1 The Four-Component Graph Block

Component	Content	Style
1 — Introduction	One sentence before the graph: what to look for, what axes represent, which curve is the worst case.	Italic, 9 pt, muted grey
2 — Figure	The matplotlib plot. Always titled inside the plot. Worst-case curve labelled. Target/limit shown as horizontal dashed line.	Full content width
3 — Caption	"Figure X.Y.Z — [description]" — full descriptive caption, not just a title. Includes what operating conditions are shown.	Italic, 8.5 pt, centred, below figure
4 — Interpretation	A technical interpretation paragraph: what the graph reveals, which curve dominates, what the trend means for the design.	Body text, 9.5 pt, justified

4.2 Live Example — Figure 4.5.3

The following shows exactly how a graph appears in Chapter 4.

Figure 4.5.3 below shows the instantaneous core loss over one half line cycle at all nine operating voltages. Focus on the 180 Vac curve (amber) — it does not peak at the line crest but shows two humps on either side, revealing the "double-hump" signature characteristic of high-line operation.

Figure 4.5.3 — Core Loss P_core(t) over Half Line Cycle — All Input Voltages

Figure 4.5.3 — Core Loss P_core(t) over Half Line Cycle — All Input Voltages

Two features stand out. First, the 180 Vac curve is the worst-case for total core dissipation despite not having the highest peak — its cycle-averaged loss of 3.04 W (max-anchored) exceeds every low-line point because the flux excursion Bac,pk is maximised near D = 0.5 (Vin = 195 V), which is reached twice per half cycle at high line. Second, the 264 Vac curve (rightmost, blue) shows almost no core loss variation across the cycle — at D ≈ 0.05 the duty cycle is so small that the core barely swings. This confirms that the thermal design point is 180 Vac, not the commonly assumed 90 Vac crest.

4.3 Graph Quality Rules

- **Every curve labelled** by input voltage on the plot itself — no legend-only identification
- **Target / limit line** as horizontal dashed on every margin plot (B_sat, ΔT_budget, etc.)
- **Worst-case curve highlighted** in amber or a distinct heavy line weight
- **Both axes labelled** with quantity name and unit
- **Plot title inside the figure** matching the figure number in the caption
- **Grid lines shown** at reduced opacity — reader can read off values

Part 5 — Table Documentation Standard

✓ AGREED Every Table Has a Name, Introduction, and Interpretation

Tables are introduced before they appear and interpreted after. No table is dropped in without context. The reader always knows what to look for in a table before they see it.

5.1 The Three-Component Table Block

Component	Content
1 — Table name	"Table X.Y.Z — [description]" in bold navy, immediately before the table. Every table has a unique number and a descriptive title.
2 — Introduction sentence	One sentence: what the table shows, what the columns represent, which row is the worst case (amber-highlighted).
3 — Table body	Assessment/verdict column on the right. Worst-case row highlighted amber. Units in column headers. Min/nom/max columns where tolerance applies.

5.2 Live Example — Table 3.5.1

Table 3.5.1 — First-Pass Loss Estimate — All 9 Operating Points (Peak-Point, Nominal AL)

Table 3.5.1 summarises the first-pass loss estimate used to qualify the core candidate. The 180 Vac row (amber) is the worst case for total loss. The "ΔT est." column confirms the thermal estimate is within the 40°C budget at all operating points. These values use the peak-point Steinmetz model — the more accurate cycle-averaged values appear in Chapter 4 Table 4.6.1.

V_in rms	P_cu @100°C (W)	P_core pk (W)	P_total (W)	ΔT est. (°C)	vs 40°C budget
90 Vac	2.59	1.97	4.56	23.7	PASS +41%
110	1.73	2.42	4.15	21.9	PASS +45%
132	1.19	2.79	3.98	21.1	PASS +47%
180 Vac ★	2.84	3.04	5.88	29.3	PASS +27%
264	1.33	1.98	3.31	18.1	PASS +55%

Table 3.5.1 — First-pass loss estimate (peak-point Steinmetz, nominal AL, 100°C copper). ★ = worst case (180 Vac). Values used for core selection only; see Table 4.6.1 for cycle-averaged results.

What Table 3.5.1 Tells Us at the Core Selection Stage

The worst-case total loss of 5.88 W at 180 Vac gives a first-pass thermal estimate of 29.3°C — a comfortable 27% margin below the 40°C budget. This confirms the selected core is thermally viable before committing to the detailed time-domain analysis in Chapter 4. The table also reveals an important design characteristic: core loss and copper loss are co-dominant at 180 Vac (3.04 W vs 2.84 W). Neither term dominates, which means neither a lower-loss core material nor a larger wire cross-section alone would significantly improve the design — both would need to be addressed simultaneously for meaningful loss reduction. Note that these are peak-point (crest) estimates; the cycle-averaged results in Chapter 4 will be 5–25% lower due to the iGSE F(D) correction and half-cycle averaging.

Part 6 — Complete Section Flow: All Standards Combined

The following shows how all five standards come together in a single report section. This is the exact sequence of elements for any major calculation section in the report.

Sequence	Element	Standard	Purpose
1	Chapter splash page	Part 1 above	Orient the reader before any content
2	Section heading — X.Y.Z	3-level decimal	Locate section in chapter hierarchy
3	CONCEPT box	Five structural elements	Why this calculation exists
4	THEORY box (optional)	Five structural elements	Mathematical derivation of governing equation
5	Equation steps (Step 1, Step 2, Step 3...)	Part 2 above	Symbolic → numerical → result
6	PITFALL box (where applicable)	Five structural elements	Most common error at this step
7	DECISION box	Five structural elements	What was selected, why, what margin it provides
8	Design Interpretation paragraph	Part 3 above	What the result means, design impact, sensitivity
9	Table introduction + table + interpretation	Part 5 above	Numerical summary of all operating points
10	Graph introduction + figure + caption + interpretation	Part 4 above	Visual evidence supporting the decision
11	INSIGHT box (where applicable)	Five structural elements	Non-obvious observation from the combined results

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Document Status: All five standards in this document are finalised and agreed. They apply to all six chapters and all appendices. The Documentation Agent will use these standards without exception.

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